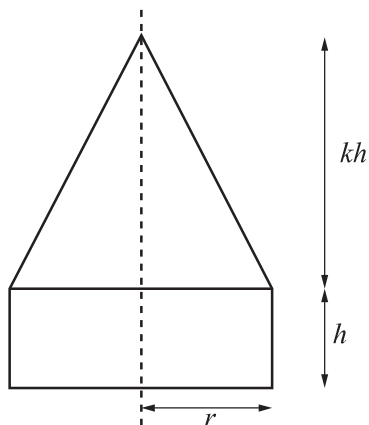




(a) Show that the distance of the centre of mass of the combined solid from the base of the cylinder is $\frac{h(k^2 + 4k + 6)}{4(3 + k)}$. [4]

[illegible]

[3]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.